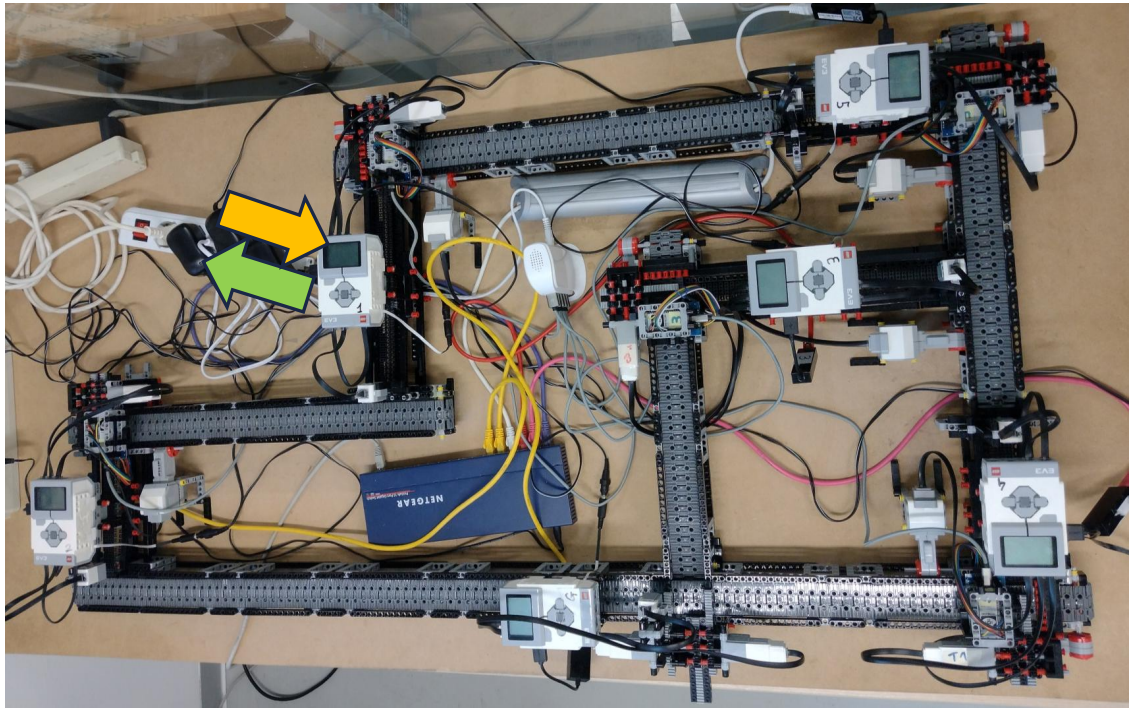
INTRODUCTION

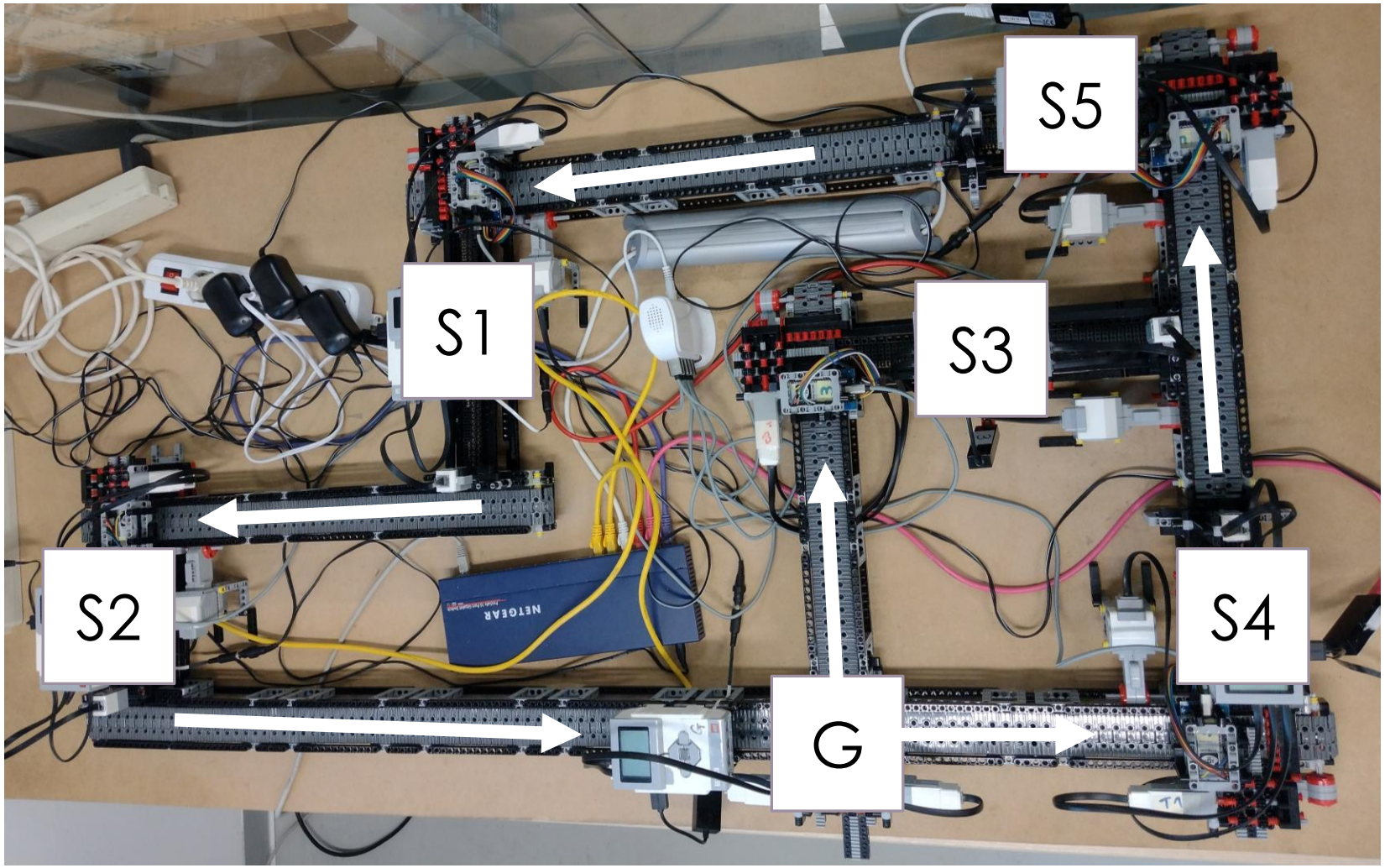
5-STATION SYSTEM

Physical System – 5 Stations

The physical system is a LEGO® Manufacturing System which can represent workstations with a single machine and an input buffer, and conveyors for part handling and holding. S1 performs also Part Load/Unload activities; thus, it is assumed that infinite raw material is available (orange arrow) and that the flow of finished goods (green arrow) exits from S1.



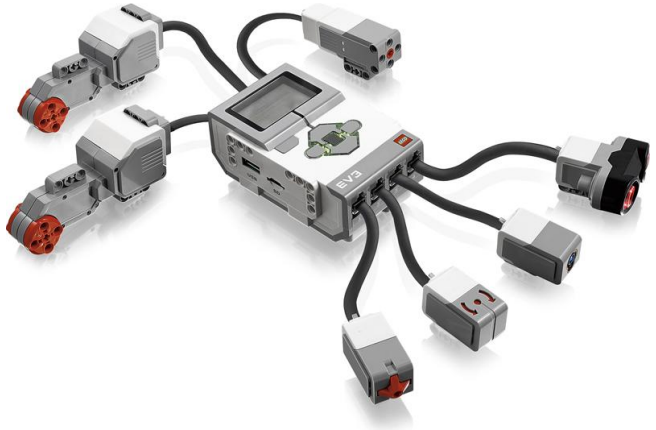
Physical System – 5 Stations

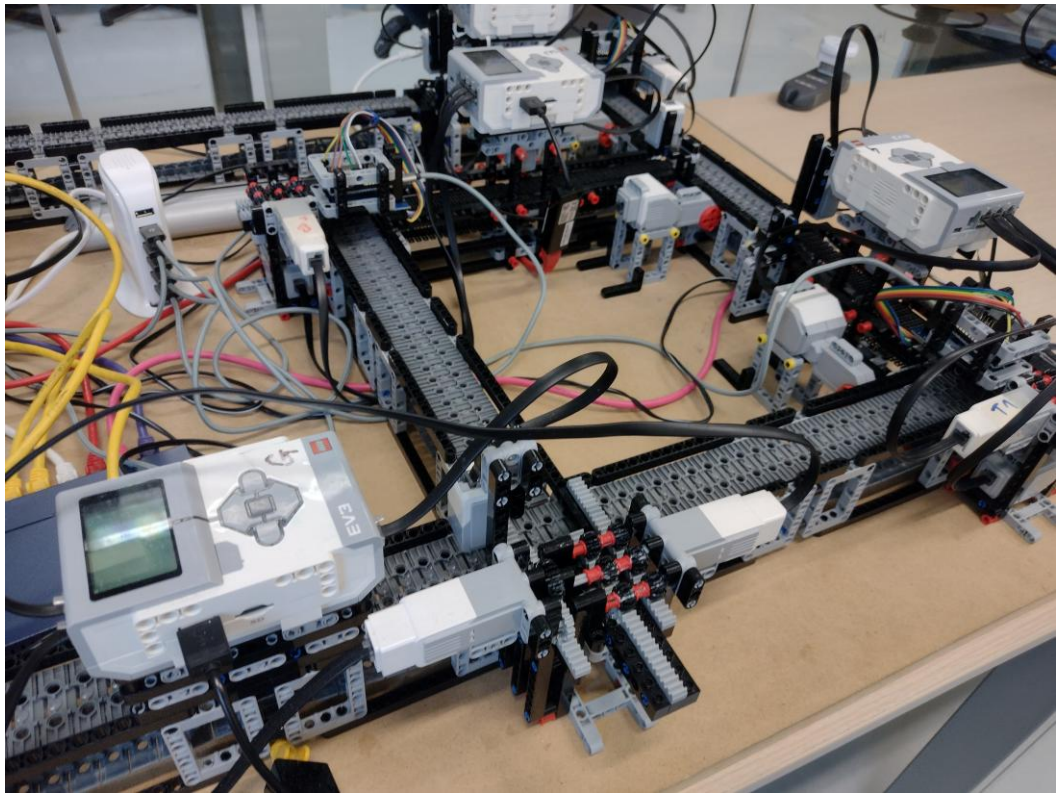


Layout Notes

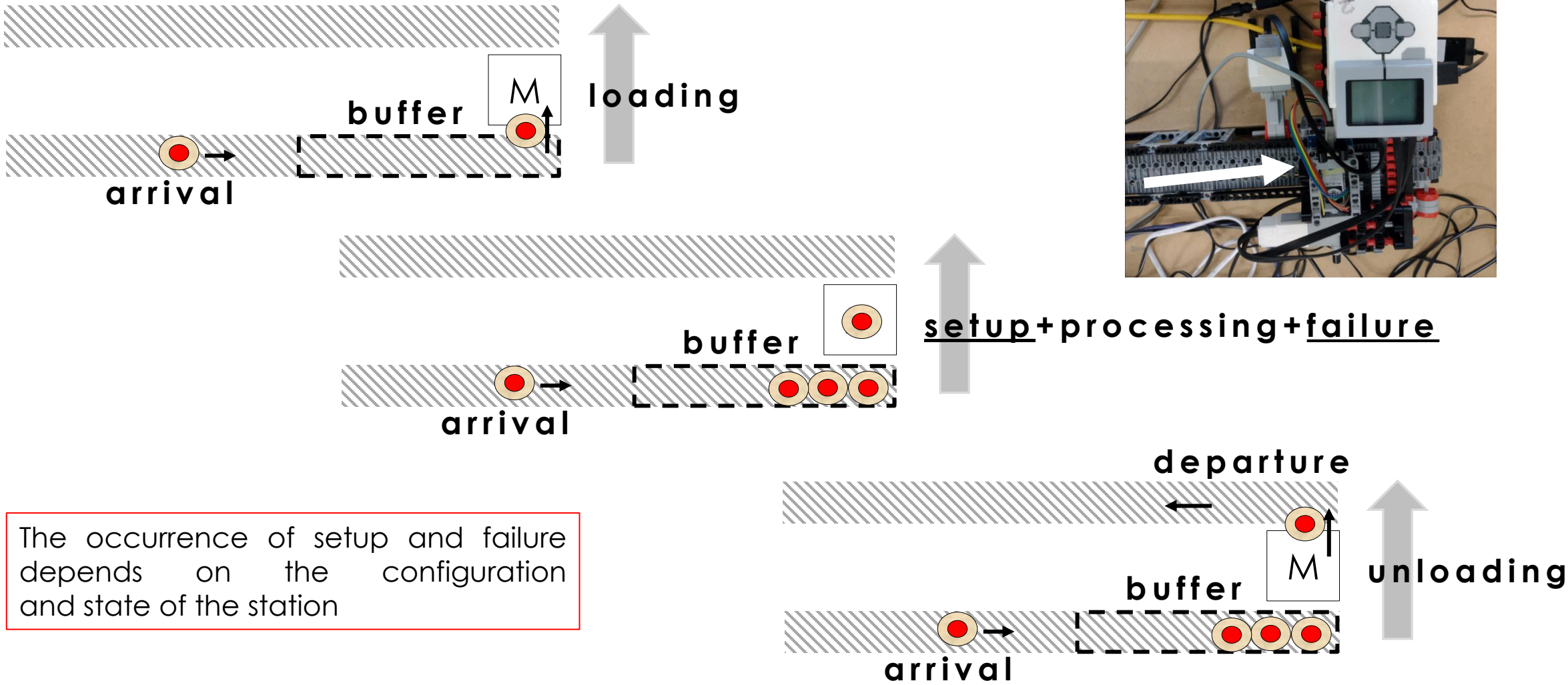
The system is a close system composed of 5 stations (S1,S2,…S5) connected in series with automatic conveyor which also serve as intermediate buffer of parts. Stations S3 and S4 works in parallel and gate "G" routes parts either to S3 or S4.

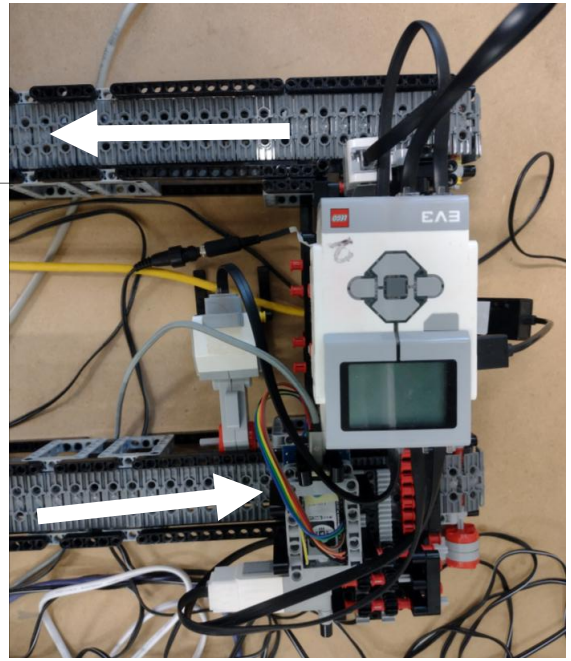
LEGO MINDSTORM Ecosystem





Workstation concept





The occurrence of setup and failure depends on the configuration and state of the station

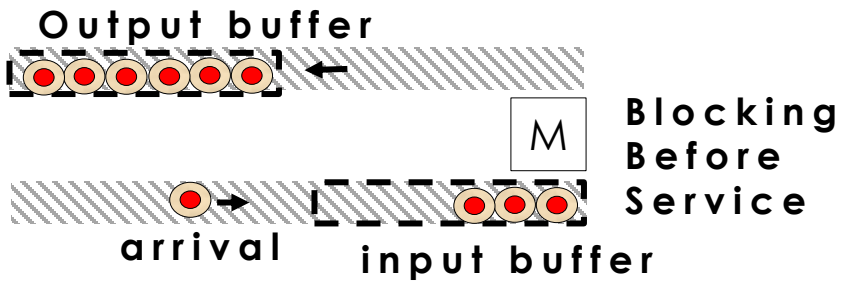
Blocking and Failures

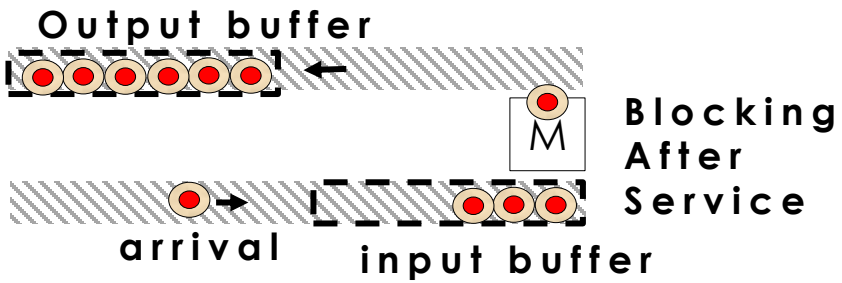
<u>Workstation state</u>:

- Red light → FAIL STATE
- Green light → READY AND IDLE
- Orange light → PROCESSING

When machine output buffer is full, **<u>the workstation blocks with BAS rule</u>**:

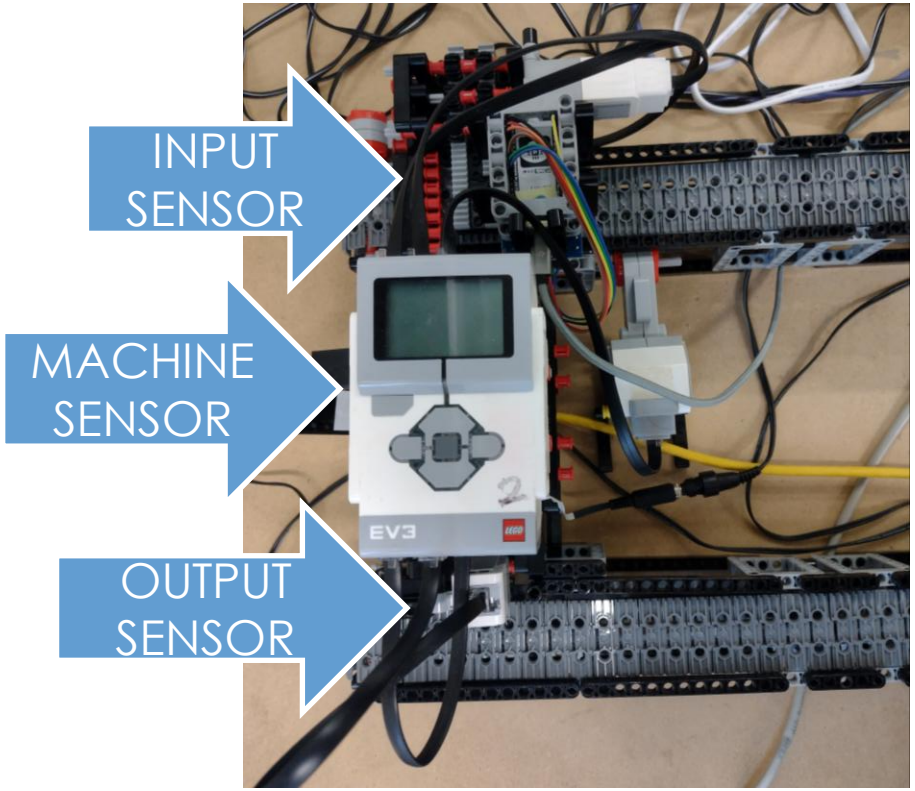
- Blocking Before Service → the machine will not load a new pallet
- **Blocking After Service** → the machine will not unload the pallet





Optical sensors

Sensors detect the presence of pallets



INPUT SENSOR

- LOAD activity

OUTPUT SENSOR

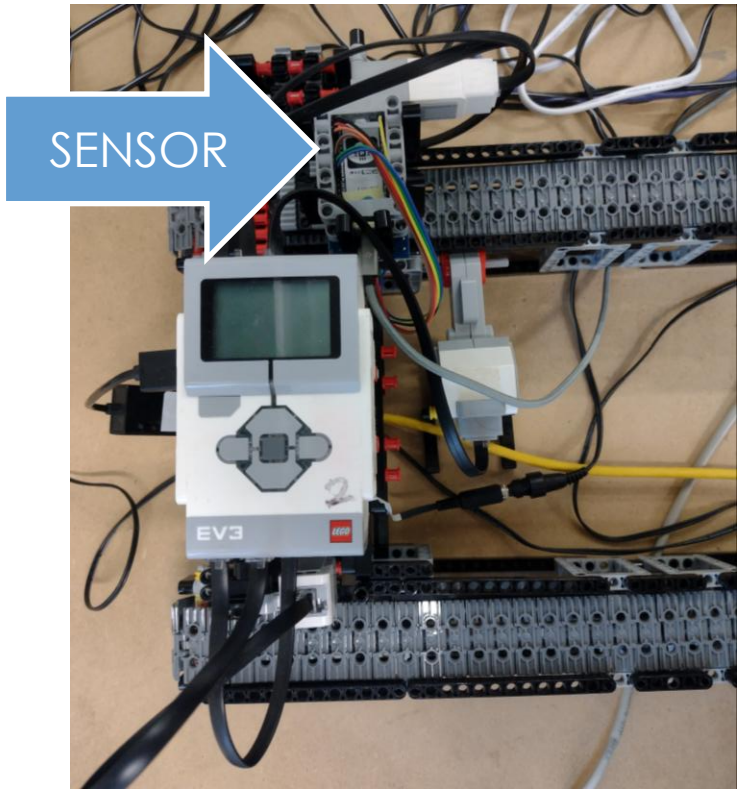
- BLOCK activity
- TRANSFER activity

MACHINE SENSOR

- PROCESS activity
- UNLOAD activity
- FAIL activity

RFID sensors

RFID sensors read and/or write information on pallets



RFID SENSOR

- PART ID

Event log of the system

Labels

Record

	A	B	C	D	E
1	time	station_id	part_id	part_type	activity
2	2025-11-18T10:51:19.345605	S1	P1	T1	LOAD
3	2025-11-18T10:51:23.466835	S1	P1	T1	PROCESS
4	2025-11-18T10:51:26.546299	S1	P1	T1	UNLOAD
5	2025-11-18T10:51:29.204537	S1	P1	T1	TRANSFER
6	2025-11-18T10:51:29.511347	S1	P7	T2	LOAD
7	2025-11-18T10:51:33.723293	S1	P7	T2	SETUP
8	2025-11-18T10:51:33.787799	S2	P1	T1	LOAD
9	2025-11-18T10:51:35.791529	S1	P7	T2	PROCESS
10	2025-11-18T10:51:37.134510	S2	P1	T1	PROCESS
11	2025-11-18T10:51:38.875545	S1	P7	T2	UNLOAD
12	2025-11-18T10:51:41.420977	S1	P7	T2	TRANSFER
13	2025-11-18T10:51:41.706779	S1	P2	T1	LOAD
14	2025-11-18T10:51:45.937571	S1	P2	T1	SETUP
15	2025-11-18T10:51:47.692970	S2	P1	T1	UNLOAD
16	2025-11-18T10:51:48.009357	S1	P2	T1	PROCESS

Sensors records 4 fields as separated by commas:

- TimeStamp
- Station ID
- Part ID
- Part Type
- Activity → depend on sensor

Configuration – Pallets

OPEN LOCATION
/g1-5s-pl/config.json

```
config.json
{
    "component_wips": {
        "S1": {"T1": 6, "T2": 6},
        "S2": {"T1": 0, "T2": 0},
        "S3": {"T1": 0, "T2": 0},
        "S4": {"T1": 0, "T2": 0},
        "S5": {"T1": 0, "T2": 0}
    },
    "log_folder": "event-logs",
    "log_columns": ["time", "station_id", "part_id", "part_type", "activity"],
    "mqtt_broker_host": "THE-FACTORY-PC1",
    "mqtt_broker_port": 1883,
    "mqtt_publication_interval": 1.0,
```

You can customize the number of circulating pallets.

Configuration – Capacity

FOR S1 OPEN LOCATION
/g1-5s-pl/s1/config.json
SIMILARLY FOR OTHER STATIONS

config.json

```
{
    "station_id" : "S1",
    "next_station_ids" : ["S2"],
    "gate_id" : null,
    "part_ids" : {
        "0xEC2B5C31":  "P1", "0xCCAB5C31":  "P2", "0xCC626531":  "P3",
        "0x9C9E5D31":  "P4", "0xACAD5C31":  "P5", "0xDCB76431":  "P6",
        "0x2C9D5D31":  "P7", "0xDC9E5D31":  "P8", "0xCCC56531":  "P9",
        "0x1C746431": "P10", "0x7CAE5C31": "P11", "0xFC0F5D31": "P12"
    },
    "part_types" : {
        "0xEC2B5C31": "T1", "0xCCAB5C31": "T1", "0xCC626531": "T1",
        "0x9C9E5D31": "T1", "0xACAD5C31": "T1", "0xDCB76431": "T1",
        "0x2C9D5D31": "T2", "0xDC9E5D31": "T2", "0xCCC56531": "T2",
        "0x1C746431": "T2", "0x7CAE5C31": "T2", "0xFC0F5D31": "T2"
    },

    "station_capacity": 5,
    "next_station_capacities" : {"S2": 6},
```

Station capacity can be customized. Note: station capacity is buffer capacity +1.

Station	Minimum capacity	Maximum capacity
S1	1	6
S2	1	6
S3	1	6
S4	1	6
S5	1	5

Note: Always verify buffer vs station capacity!

Configuration – Setup and processing

FOR S1 OPEN LOCATION
/g1-5s-pl/s1/config.json
SIMILARLY FOR OTHER STATIONS

config.json

```
"station_capacity": 5,
"next_station_capacities" : {"S2": 6},
"setup_time_distributions": {
    "T1": {
        "T1": null,
        "T2": {
            "type": "DETERMINISTIC",
            "parameters": [2.0]
        }
    },
    "T2": {
        "T1": {
            "type": "DETERMINISTIC",
            "parameters": [2.0]
        },
        "T2": null
    }
},
"processing_time_distributions": {
    "T1": {
        "type": "DETERMINISTIC",
        "parameters": [3.0]
    },
    "T2": {
        "type": "DETERMINISTIC",
        "parameters": [3.0]
    }
},
```

Setup and processing time distributions can be customized.

type	parameters
DETERMINISTIC	[value]
UNIFORM	[min, max]
TRIANGULAR	[high, low, mode]
BETA	[alpha, beta]
EXPONENTIAL	[rate]
GAMMA	[shape, scale]
LOG_NORMAL	[mean, std] (of the normal distribution)
NORMAL	[mean, std]
VON_MISES	[mean, 1/variance]
PARETO	[shape]
WEIBULL	[scale, shape]
REPLAYER	[path, delimiter]

Configuration – Failure and routing

FOR S1 OPEN LOCATION
/g1-5s-pl/s1/config.json
SIMILARLY FOR OTHER STATIONS

config.json

```
"failure_probability": 0.1,
"repair_time_distribution": {
    "type": "DETERMINISTIC",
    "parameters": [8.0]
},
"routing_policies": {
    "T1": null,
    "T2": null
},
"blocking_policy": {
    "type": "BLOCK_AFTER_SERVICE",
    "parameters": null
},
```

Station failures can be customized in terms of failure probability and repair time distribution.

FOR S2 OPEN LOCATION
/g1-5s-pl/s2/config.json

config.json

```
"failure_probability": 0.0,
"repair_time_distribution": null,
"routing_policies": {
    "T1": {
        "type": "PROBABILITY",
        "parameters": {"S3": 0.5, "S4": 0.5}
    },
    "T2": {
        "type": "PROBABILITY",
        "parameters": {"S3": 0.5, "S4": 0.5}
    }
},
"blocking_policy": {
    "type": "BLOCK_AFTER_SERVICE",
    "parameters": null
},
```

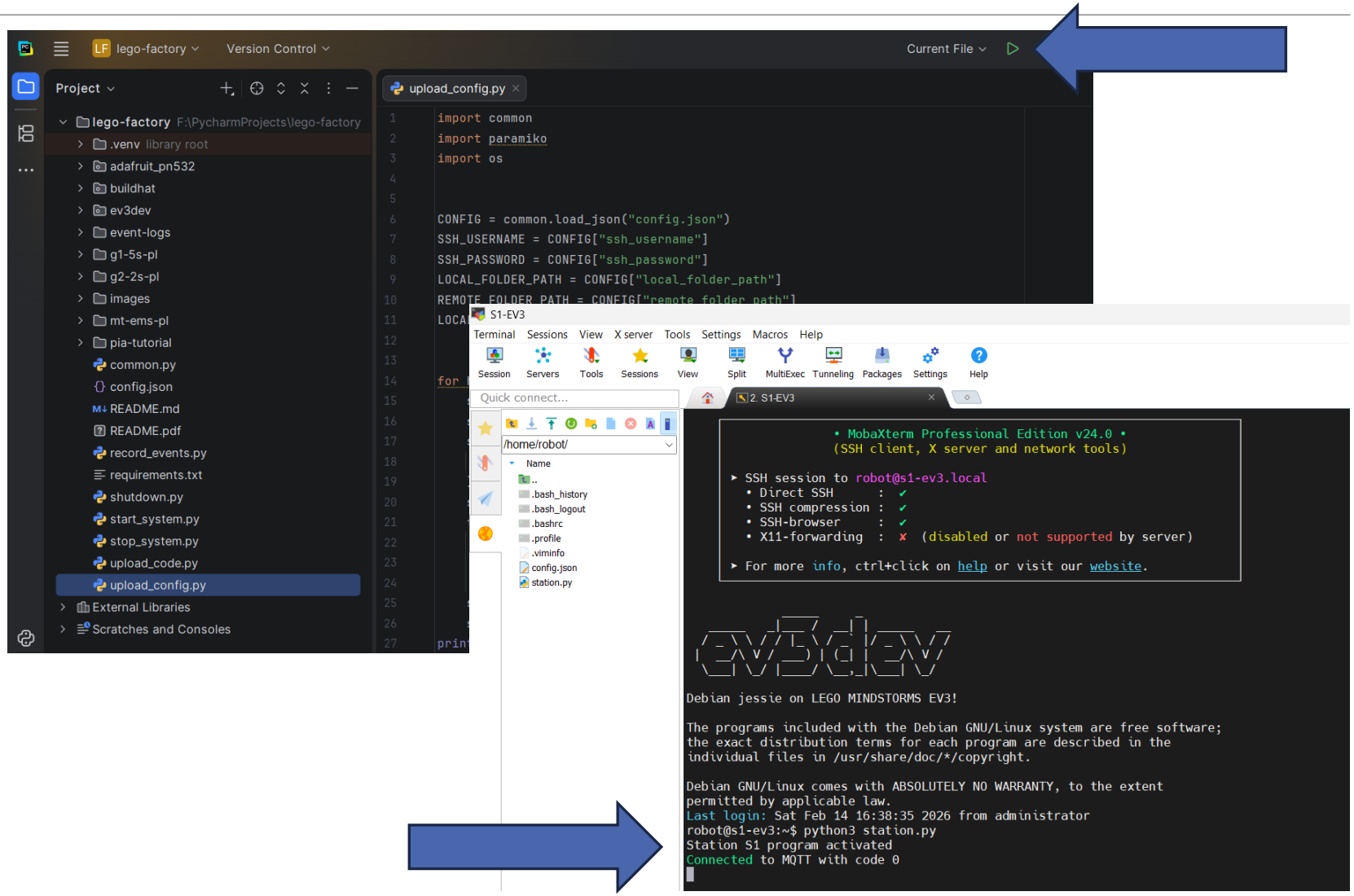
Pallet routing (allocation) is probabilistic, i.e. a part is allocated to S3 with probability 0.5, otherwise to S4.
You can customize probability.

Configuration

AFTER MODIFYING THE CONFIG FILE OF A STATION, FOR EXAMPLE **/g1-5s-pl/s1/config.json**, UPLOAD IT TO THE CORRESPONDING CONTROLLER, FOR EXAMPLE **S1-EV3**, BY RUNNING THE **upload_config.py** SCRIPT

IF THE STATION PROGRAME, FOR EXAMPLE **station.py**, HAS ALREADY BEEN RUNNING, STOP AND RESTART THE PROGRAM SO THAT THE NEW CONFIG FILE CAN TAKE EFFECT

ALWAYS UPLOAD YOUR OWN SET OF STATION CONFIG FILES BEFORE CARRYING OUT EXPERIMENTS



Configuration

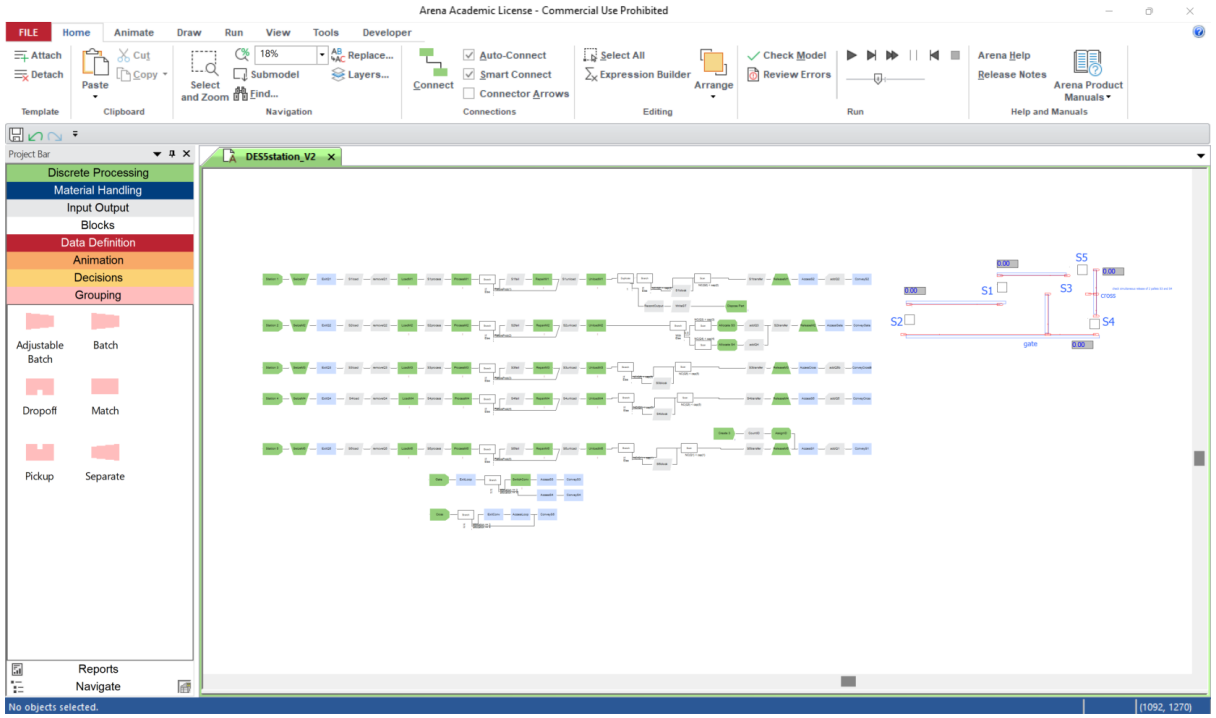
The following system parameters should not be changed:

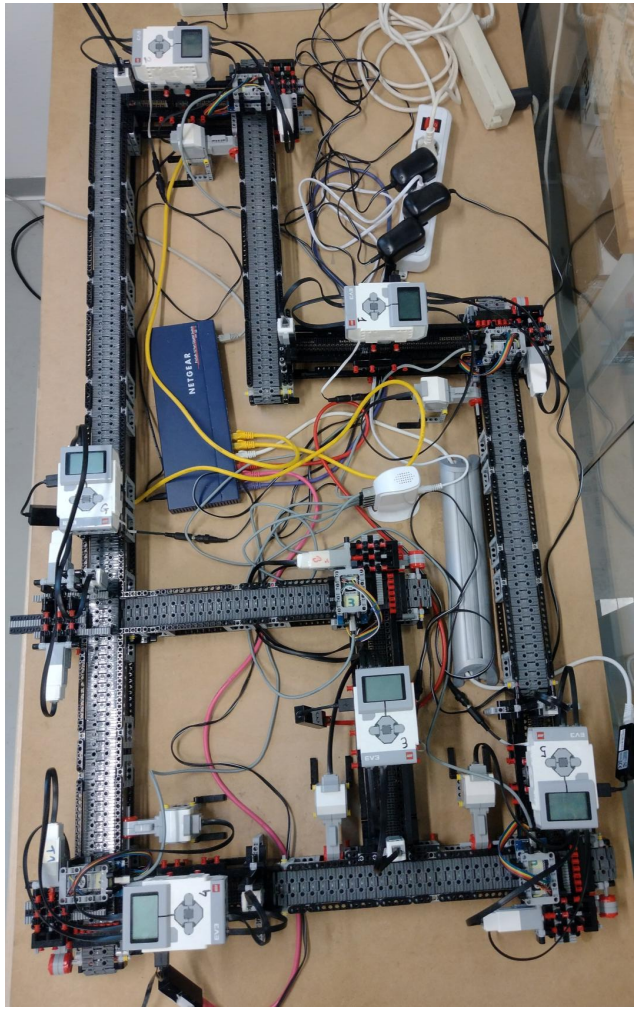
- **Keep the recommended conveyor speed, motor parameters and delays, sensor threshold and other parameters for system automation and communication infrastructure;**
- We have a maximum of 12 available pallets, any increase should be discussed;
- Buffer capacity is related to conveyor length. Despite the system may accommodate more pallets, their weight can damage the conveyor. Therefore, the station capacity must be selected within the provided range:

Station	Minimum capacity	Maximum capacity
S1	1	6
S2	1	6
S3	1	6
S4	1	6
S5	1	5

DES model

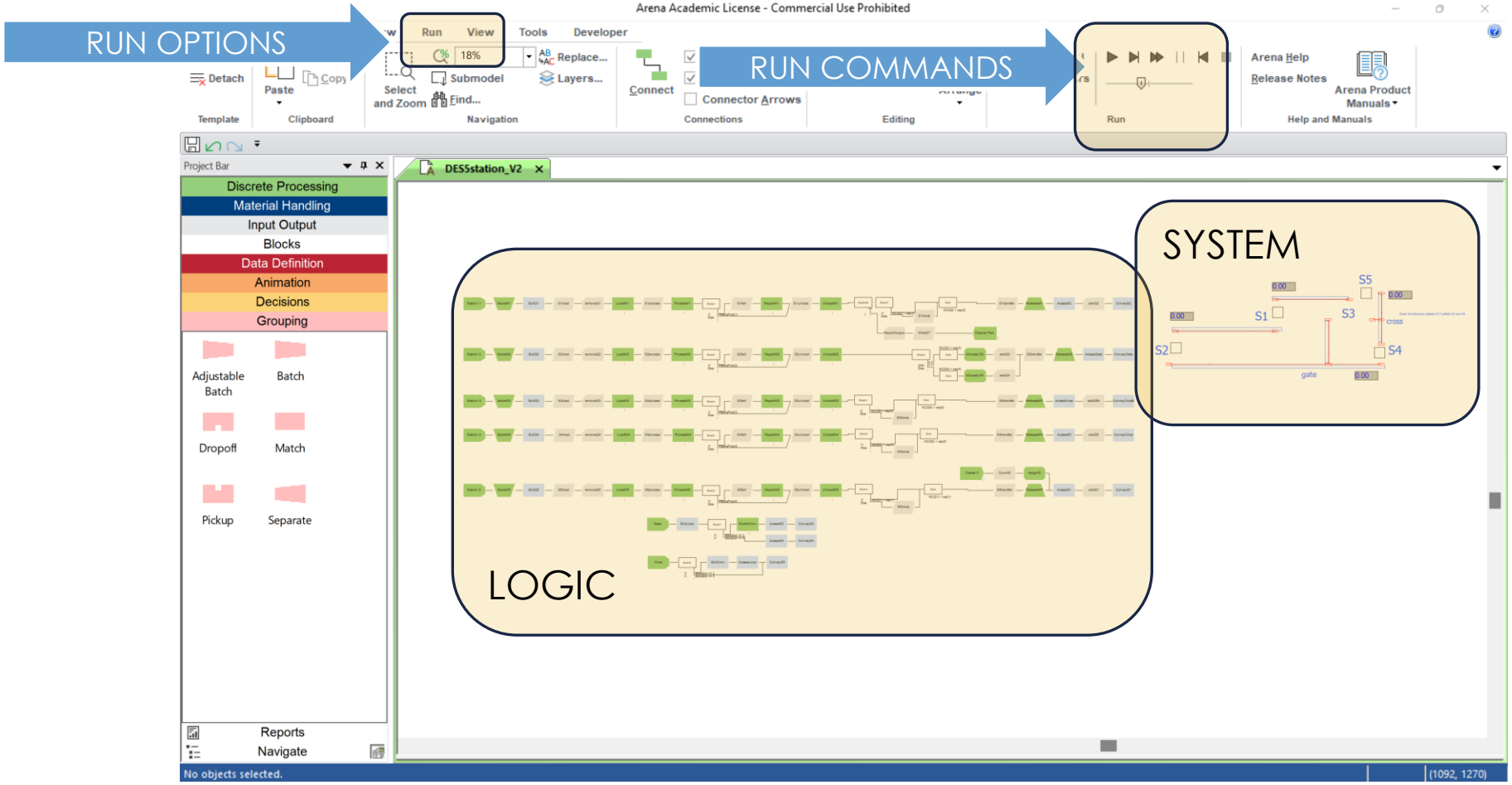
The Discrete Event Simulation (DES) model mimics the behavior of the real system and provides event-log. You need ARENA to execute the model ("STUDENT" licence).



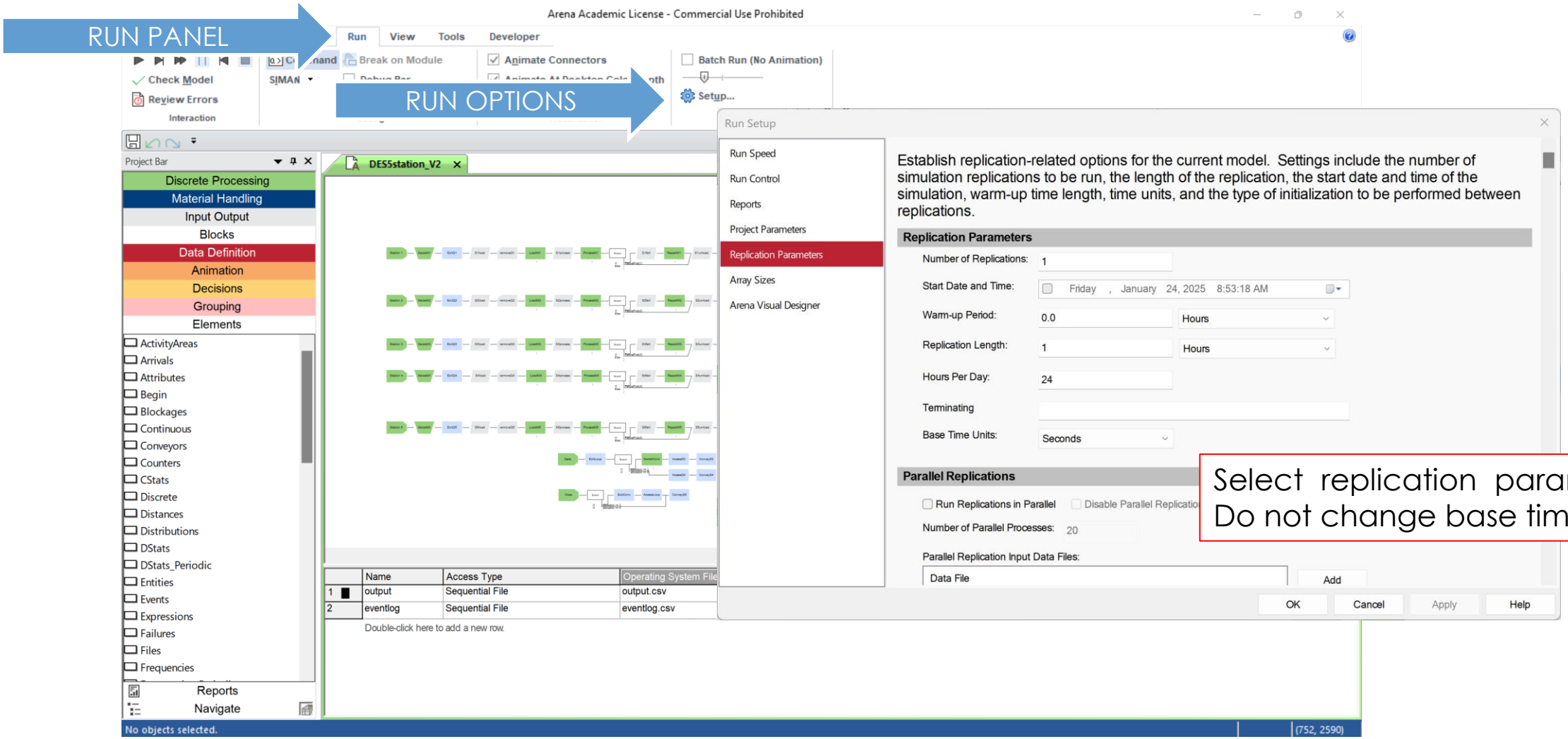


https://www.rockwellautomation.com/it-it/products/software/arena-simulation.html

DES model

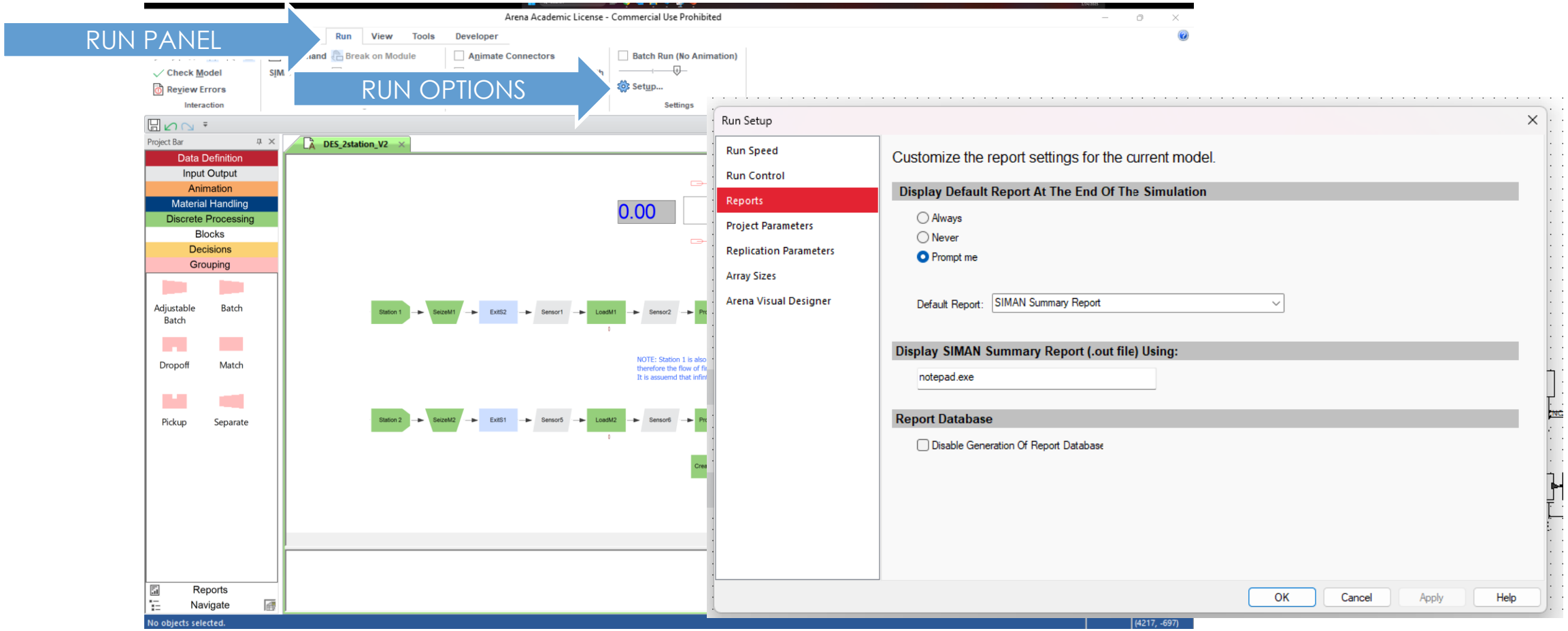

RUN OPTIONS
RUN COMMANDS
SYSTEM
LOGIC
Arena Academic License - Commercial Use Prohibited
Run
View
Tools
Developer
Detach
Paste
Copy
Template
Clipboard
Select and Zoom
Submodel
Find...
Replace...
Layers...
Navigation
Connect
Connector Arrows
Connections
Editing
Run
Arena Help
Release Notes
Arena Product Manuals
Help and Manuals
Project Bar
Discrete Processing
Material Handling
Input Output
Blocks
Data Definition
Animation
Decisions
Grouping
Adjustable Batch
Batch
Dropoff
Match
Pickup
Separate
Reports
Navigate
DES5station_V2
S1
S2
S3
S4
S5
No objects selected.
(1092, 1270)

DES model – RUN OPTIONS



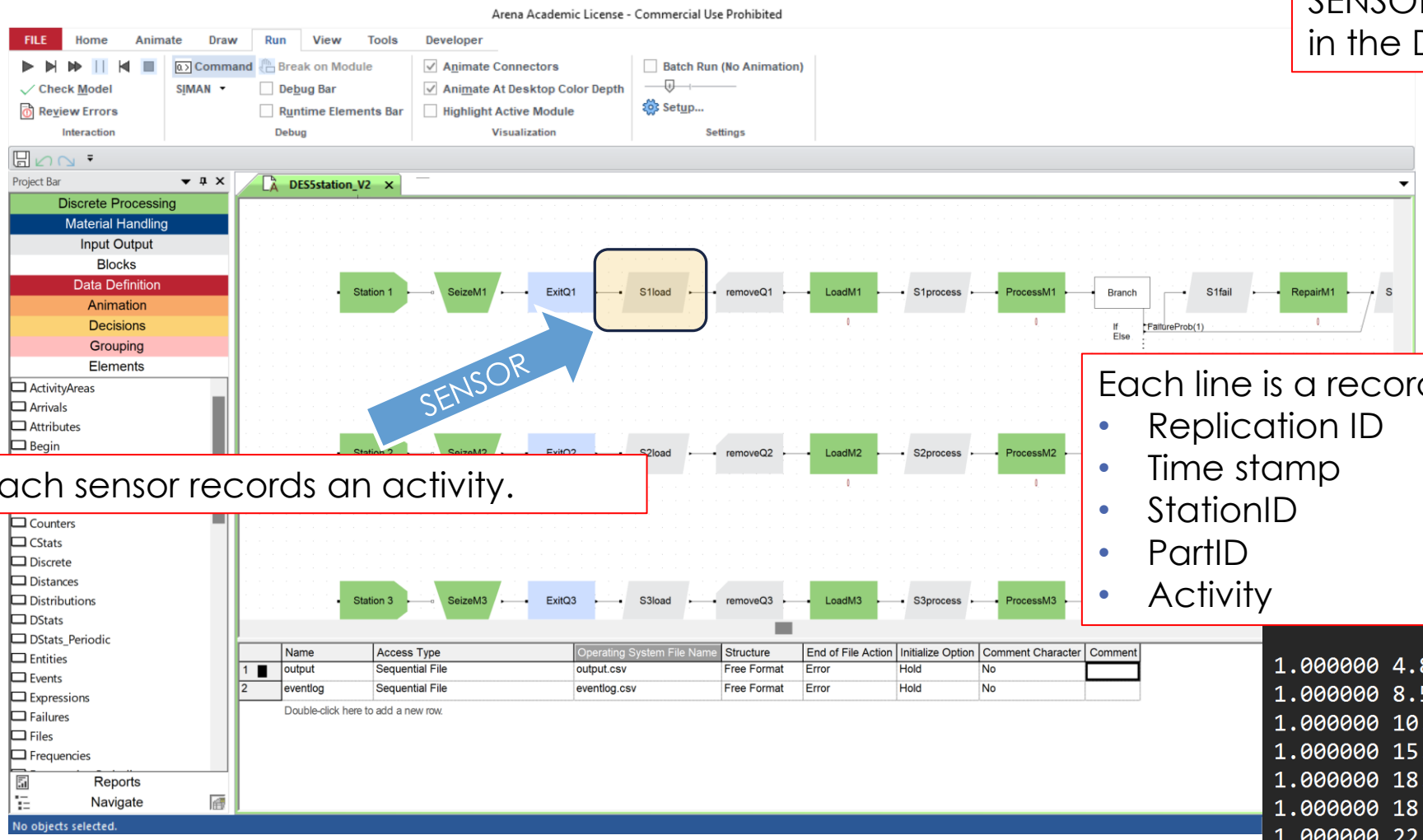
Select replication parameters.
Do not change base time units.

DES model – REPORT



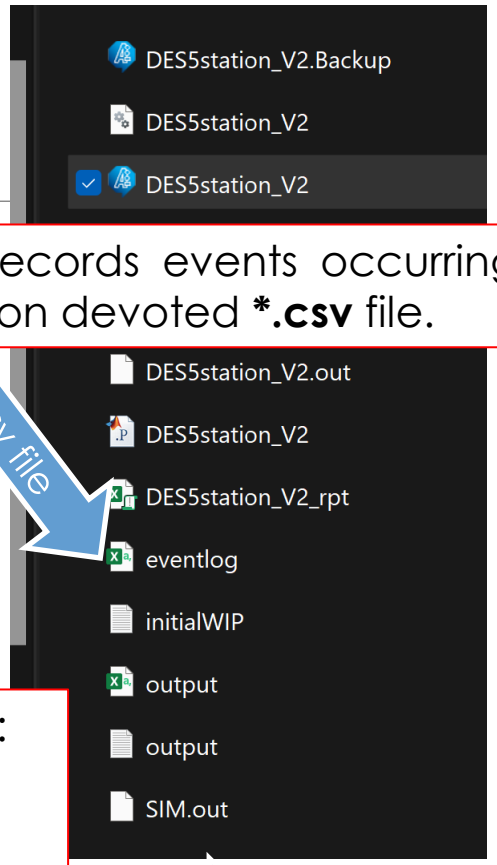
DES model – Sensors

SENSORS records events occurring in the DES on devoted ***.csv** file.



SENSOR

Each sensor records an activity.



csv file

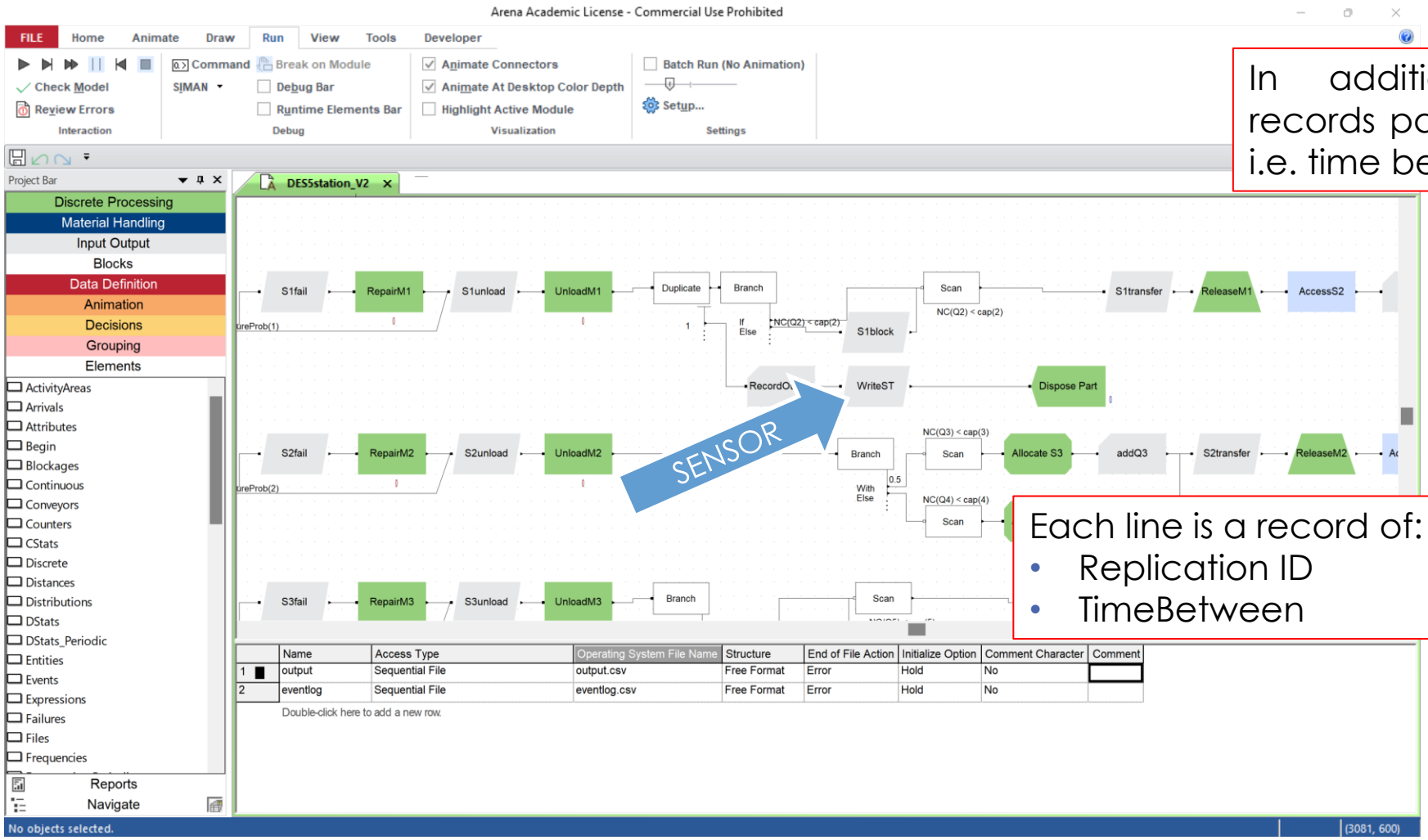
Each line is a record of:

- Replication ID
- Time stamp
- StationID
- PartID
- Activity

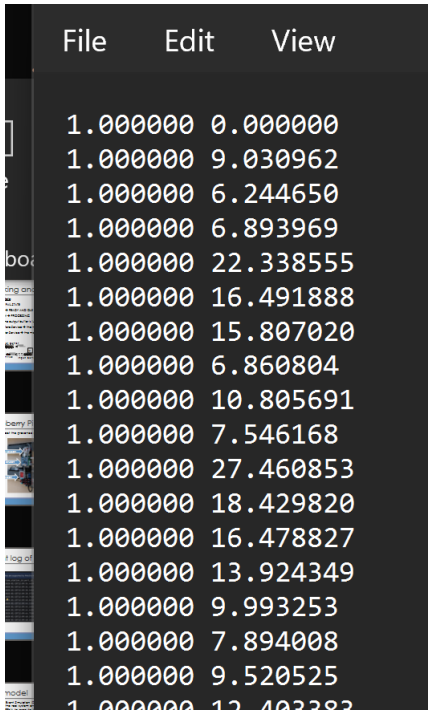
eventlog.csv

```
1.000000 4.800000 1.000000 1.000000 LOAD
1.000000 8.500000 1.000000 1.000000 PROCESS
1.000000 10.859317 1.000000 1.000000 FAIL
1.000000 15.875160 1.000000 1.000000 UNLOAD
1.000000 18.375160 1.000000 1.000000 TRANSFER
1.000000 18.375160 1.000000 2.000000 LOAD
1.000000 22.075160 1.000000 2.000000 PROCESS
1.000000 22.500000 2.000000 1.000000 LOAD
1.000000 24.133557 1.000000 2.000000 FAIL
```

DES model – Sensors



In addition, Sensor "WriteST" records parts interdeparture times, i.e. time between two exits.

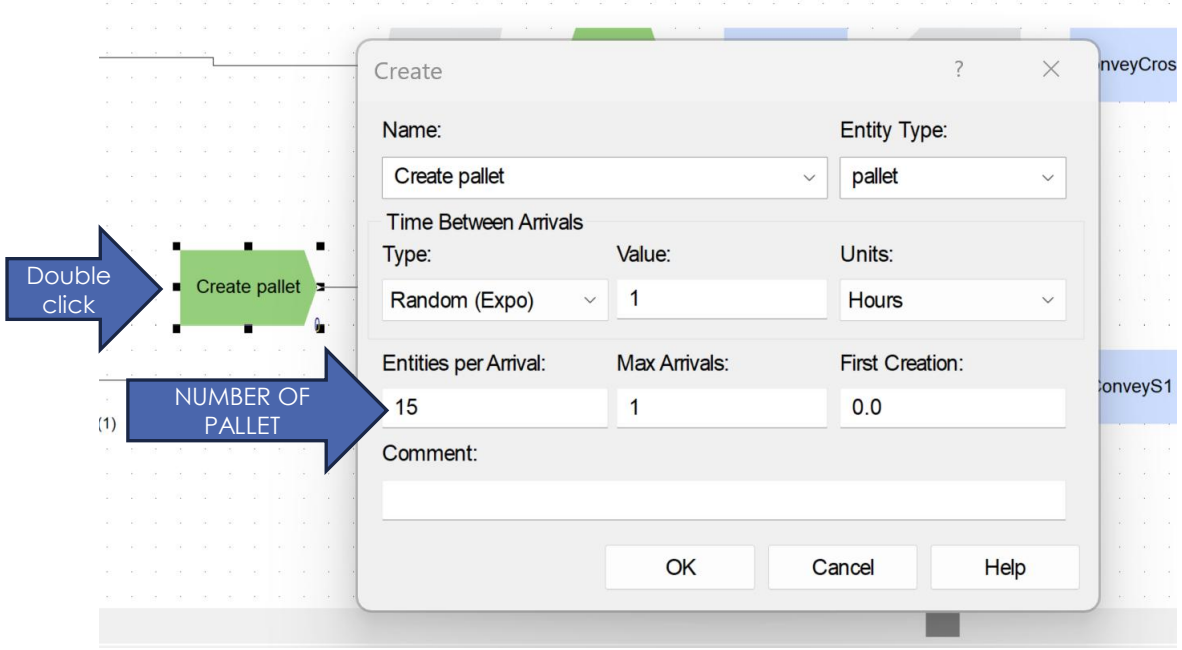


Each line is a record of:

- Replication ID
- TimeBetween

DES model - Configuration

System parameters can be configured in ARENA environment.

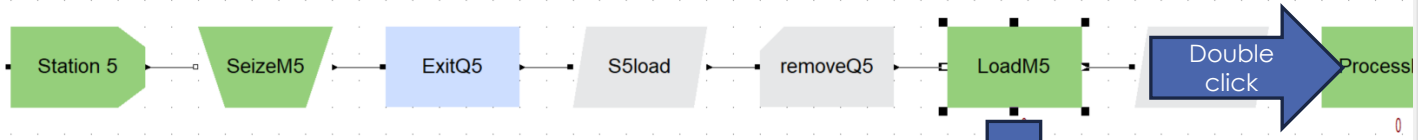


You can customize each block by double-click on the block.
For example, you can change the **number of pallets** in the CREATE block.

DIFFERENTLY FROM THE REAL SYSTEM, "S1" is the initial location of all the pallets.

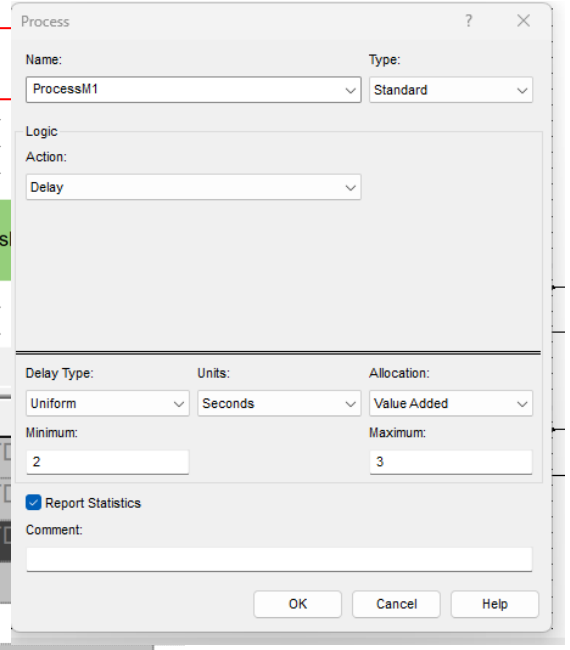
DES model - Configuration

System parameters can be configured in ARENA environment.



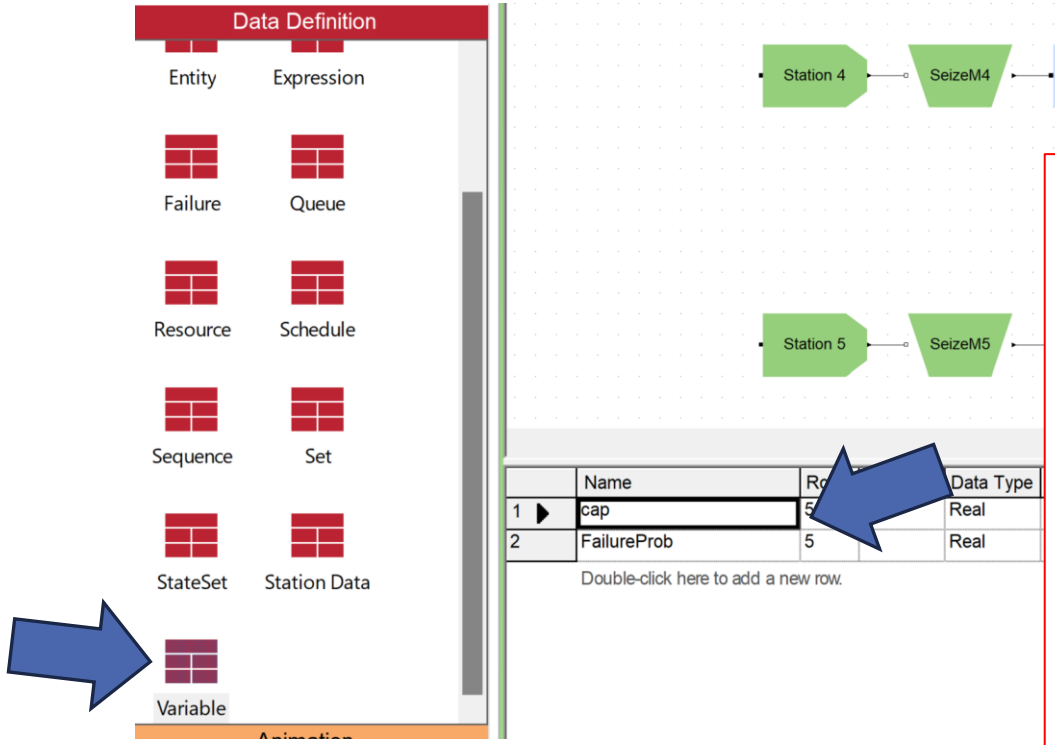
Select any process block and customize the list below. You can change the time distributions.

Name	Type	Action							Expression
LoadM3	Standard	Delay	Constant	Seconds	Value Added	.5	.6	1.5	PT1 + EXPO(STD
LoadM4	Standard	Delay	Constant	Seconds	Value Added	.5	3	1.5	PT1 + EXPO(STD
LoadM5	Standard	Delay	Constant	Seconds	Value Added	.5	.9	1.5	PT1 + EXPO(STD
ProcessM1	Standard	Delay	Uniform	Seconds	Value Added	2	10	3	10
ProcessM2	Standard	Delay	Expression	Seconds	Value Added	.5	10	1.5	LOGN(6.5,4.7)
ProcessM3	Standard	Delay	Constant	Seconds	Value Added	.5	3	1.5	10
ProcessM4	Standard	Delay	Expression	Seconds	Value Added	.5	10	1.5	EXPO(9.5)
ProcessM5	Standard	Delay	Expression	Seconds	Value Added	.5	10	1.5	TRIA(11,12.5,14)
RepairM1	Standard	Delay	Expression	Seconds	Value Added	2	10	3	WEIB(12,1.5)
RepairM2	Standard	Delay	Expression	Seconds	Value Added	2	10	3	WEIB(12,1.5)
RepairM3	Standard	Delay	Expression	Seconds	Value Added	2	10	3	WEIB(12,1.5)
RepairM4	Standard	Delay	Expression	Seconds	Value Added	2	10	3	WEIB(12,1.5)



DES model - Configuration

System parameters can be configured in ARENA environment.



Project Bar
→ Data definition
→ Variable
→ use **FailureProb** to change machine failure occurrence
→ use **cap** to change buffer capacity

Buffer	Variable	Minimum capacity	Maximum capacity
B1	cap(1)	1	5
B2	cap(2)	1	5
B3	cap(3)	1	5
B4	cap(4)	1	5
B5	cap(5)	1	4

Note: Always verify buffer vs station capacity!

DES model - Configuration

The following system parameters should not be changed:

- Keep the recommended conveyor speed;
- Do not change loading and unloading times, being related to the machine inner-conveyor speed;
- In the DES you do not have a limit of pallets.
- In the DES, Buffer capacity is only related to conveyor length. Therefore, the buffer capacity must be selected within the provided range:

Buffer	Minimum capacity	Maximum capacity
B1	1	5
B2	1	5
B3	1	5
B4	1	5
B5	1	4